

Day 4: Component-Specific Troubleshooting Practical

Task 1 : Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Procedure

Step 1 – Shut Down the Computer

I completely shut down the computer and disconnected the power cable/charger. This was done to prevent electrical damage while handling the RAM.

Step 2 – Open the Computer

I opened the computer case/back cover carefully using the appropriate screwdriver where required. Before touching the internal components, I discharged static electricity by touching a grounded metal surface.

Step 3 – Locate the RAM

I located the **RAM module and its RAM slot (DIMM/SO-DIMM slot)** on the motherboard. The RAM is held in place by retaining clips at the sides of the slot.

Step 4 – Remove the RAM

I released the retaining clips on both sides of the RAM slot and carefully removed the RAM module by holding its edges. I avoided touching the gold contacts on the RAM.

Step 5 – Inspect the RAM and Slot

I checked the RAM module and slot for:

- Dust or dirt
- Physical damage
- Damaged or bent contacts
- Loose components
- Any visible signs of overheating or corrosion

Observation: No physical damage was observed.

Step 6 – Re-insert the RAM

I aligned the notch on the RAM module with the corresponding key in the RAM slot. I then pressed the RAM module down firmly and evenly until the retaining clips locked it into position.

Step 7 – Close the Computer

After confirming that the RAM was properly seated, I closed the computer case/back cover and reconnected the power cable/charger.

Step 8 – Test the Computer

I powered on the computer and checked whether it booted normally. I also checked whether the system detected the installed memory correctly.

Observation: The computer booted normally and no unusual beep or error was observed.

Issues Observed

No physical damage, broken clips, or visible problems were observed during the RAM reseating process. The RAM was successfully removed and re-installed, and the computer operated normally after the procedure.

Conclusion

The RAM reseating procedure was completed successfully. Reseating RAM is a useful troubleshooting step when a computer experiences problems such as **no display, boot failure, random restarts, or memory-related POST errors**. Proper alignment and secure installation of the RAM module are important for reliable system operation.

Safety Precautions

- Always shut down and disconnect power before handling RAM.
- Hold RAM by its **edges** and avoid touching the gold contacts.
- Avoid static electricity by using proper ESD precautions.
- Never force the RAM into the slot; align the notch correctly before installation.

Task 2 : Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

Step	Troubleshooting Check	What to Check / Action	Result to Confirm
1. Identify the symptoms	Observe the problem carefully.	Check for no display, screen flickering, graphical artifacts, crashes, black screen, or sudden FPS drops.	Note the exact symptom and when it occurs.
2. Check the monitor and display cable	Inspect the HDMI/DisplayPort cable and connections.	Make sure the cable is firmly connected to the GPU's display output , not accidentally connected to the motherboard's video output. Try another cable/port if available.	Display should appear normally without flickering.
3. Check the GPU physical connection	Power off the PC and inspect the graphics card.	Make sure the GPU is properly seated in the PCIe slot and its retaining bracket is secure.	GPU should be firmly installed without being loose.
4. Check GPU power connectors	Inspect the PCIe power cables connected to the GPU.	Ensure all required 6-pin/8-pin or appropriate GPU power connectors are fully inserted. Check for loose or damaged cables.	GPU receives proper power and the system starts normally.
5. Check the GPU drivers	Check the graphics adapter in Windows Device Manager.	Look for warning symbols or driver errors. Install/update the correct GPU driver from the GPU	GPU should be detected correctly without driver errors.

		manufacturer's official source if necessary.	
6. Check GPU temperature and cooling	Inspect GPU fans and cooling system.	Make sure the fans can spin freely and the heatsink/air vents are not heavily blocked by dust. Excessive temperature can cause crashes or shutdowns.	GPU should remain at a reasonable temperature during normal use.
7. Test the GPU under load	Run a normal game or a suitable GPU stress/benchmark test.	Monitor whether the previous artifacts, crashes, black screens, or performance problems return.	System should remain stable during the test.
8. Confirm the fix	Use the PC normally and test multiple graphics workloads.	Run the affected game and check display quality, stability, temperatures, and performance.	If the original problem no longer occurs, the troubleshooting can be considered successful.

- ❖ A faulty-GPU investigation should start with simple checks such as **identifying the display symptom and checking cables**, followed by inspecting the GPU's physical and power connections. Driver and cooling checks should then be performed before testing the GPU under load. The fix is confirmed when the PC operates normally without the original **display problems, crashes, artifacts, or instability**.

Task 3 : Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Procedure

Step 1 – Check the Storage Device Before Disconnecting

The computer was started normally and the BIOS/UEFI Setup Utility was opened. The storage information was checked to confirm that the SSD/HDD was detected.

Observation: The storage device was detected normally in the BIOS, and the system was able to boot from the installed operating system.

Step 2 – Shut Down the Computer

The computer was completely shut down and the power cable was disconnected. This was done before handling the internal storage device.

Step 3 – Disconnect the Storage Device

For a SATA drive, the **SATA data cable and power cable** were disconnected.

For an NVMe drive, the computer was powered off and the NVMe drive was carefully removed from its M.2 slot.

Step 4 – Start the Computer Without the Drive

The computer was powered on without the storage device connected.

Possible BIOS indication:

- The storage drive was no longer listed in the BIOS/UEFI storage information.
- If the disconnected drive contained the operating system, the PC could display an error such as **“No Bootable Device,” “Boot Device Not Found,”** or **“No Operating System Found.”**

The exact message depends on the motherboard and BIOS/UEFI implementation.

Step 5 – Reconnect the Storage Device

The computer was switched off again. The storage device was reconnected securely.

For a SATA drive, both the data and power connections were checked. For an NVMe drive, the drive was correctly inserted into the M.2 slot and secured.

Step 6 – Check the BIOS Again

The computer was started and the BIOS/UEFI Setup Utility was opened.

Observation: The storage device appeared again in the BIOS storage/device list after it was properly reconnected.

Step 7 – Boot the Operating System

The system was allowed to boot normally.

Observation: The operating system loaded successfully after the storage device was detected again.

Before and After Comparison

Condition	BIOS/Boot Observation
Before disconnecting	SSD/HDD detected in BIOS and operating system booted normally.
Drive disconnected	Drive not detected in BIOS; if it was the boot drive, a No Bootable Device/Boot Device Not Found type of error could appear.
After reconnecting	Drive detected again in BIOS and the operating system booted normally.

Conclusion

Disconnecting a storage device can cause the BIOS/UEFI to stop detecting it. If the disconnected drive contains the operating system, the computer may fail to boot and display a **No Bootable Device** or similar error. After reconnecting the drive securely, the BIOS should detect the storage device again and the system should normally be able to boot.

Task 4 : Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Component I Would Check First: RAM

I would check the **RAM first** because faulty, loose, or improperly seated RAM can cause **random restarts, system freezes, crashes, and intermittent boot problems**. RAM is also relatively easy to inspect and test compared with replacing or troubleshooting the GPU or storage drive.

Diagnostic Steps for RAM

- 1. Shut down the PC and disconnect the power.**
 - Ensure the system is completely powered off before opening the case.
- 2. Inspect the RAM physically.**
 - Check for dust, visible damage, or improperly seated modules.
 - Make sure the RAM retaining clips are properly locked.
- 3. Reseat the RAM.**
 - Remove the RAM module carefully by releasing the side clips.
 - Reinsert it firmly until both retaining clips lock into place.
- 4. Test one RAM module at a time.**
 - If multiple RAM sticks are installed, remove all but one.
 - Boot the PC and observe whether random restarts continue.
 - Repeat the test with the other RAM module.
- 5. Try different RAM slots.**
 - If the problem continues, test the same RAM module in another compatible slot.
 - This can help identify a potentially faulty DIMM slot.
- 6. Check for POST errors.**
 - Restart the PC and listen for unusual beep patterns or check motherboard diagnostic LEDs.
 - Memory-related POST errors can indicate a RAM or memory-slot problem.

7. Confirm stability.

- Use the computer normally or run a memory test if software diagnostics are permitted.
- If the random restarts stop after replacing/reseating a particular RAM module, that provides evidence that the memory was involved.

Why RAM Before GPU or Storage?

- **RAM:** Can directly cause random crashes and restarts and is easy to reseat/test.
- **GPU:** More commonly associated with **no display, graphical artifacts, flickering, or crashes during graphics-heavy workloads.**
- **Storage:** More commonly causes **slow performance, boot errors, missing files, or operating-system errors** rather than immediate random restarts.

Conclusion

For a PC that randomly restarts, I would **start with RAM**, checking its physical connection and testing the modules individually. If the RAM checks out, I would then investigate other possible causes such as the **GPU, storage, PSU, CPU temperature, or motherboard.**